

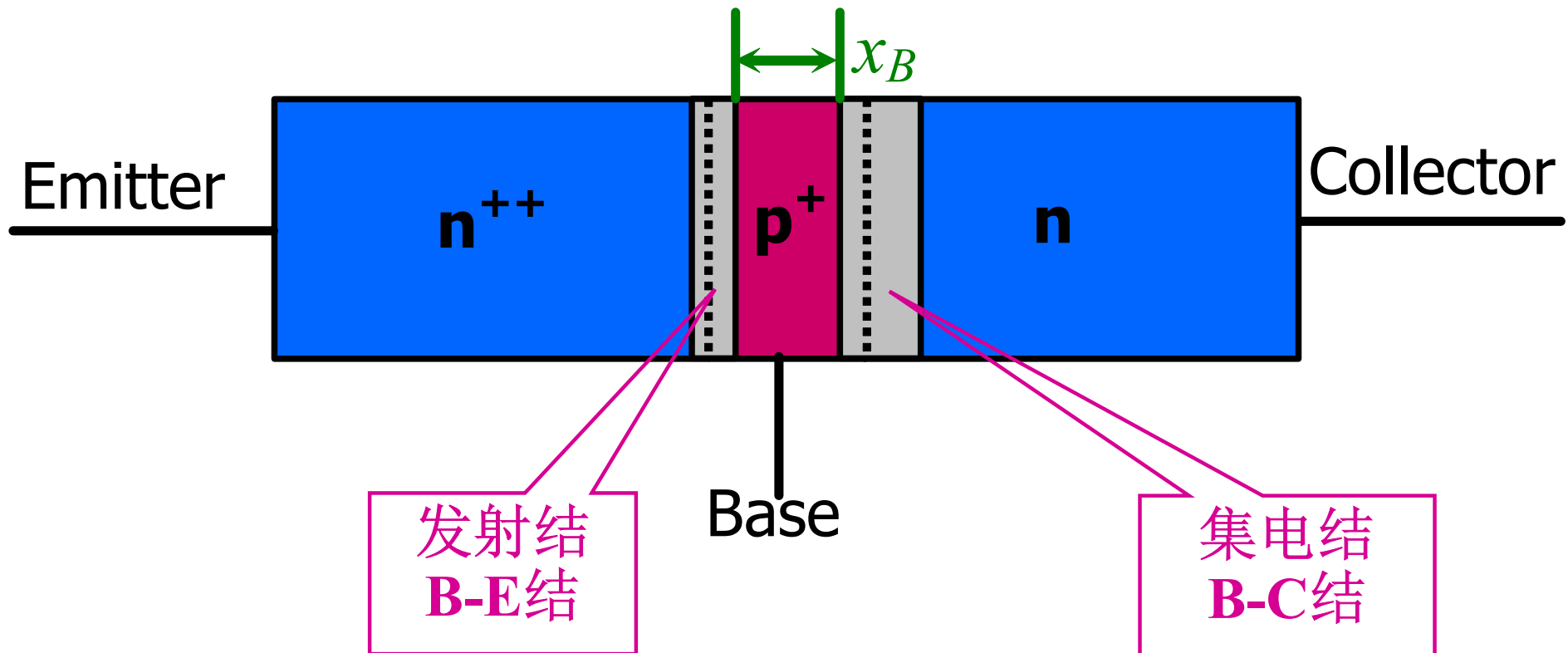
期末复习

第四章小结

1. BJT是由靠得很近的两个pn结构成的半导体器件
发射区掺杂浓度最高，基区次之，而集电区的掺杂浓度则最低

电中性基区宽度 $\ll$ 基区少子扩散长度， 即 $x_B \ll L_B$

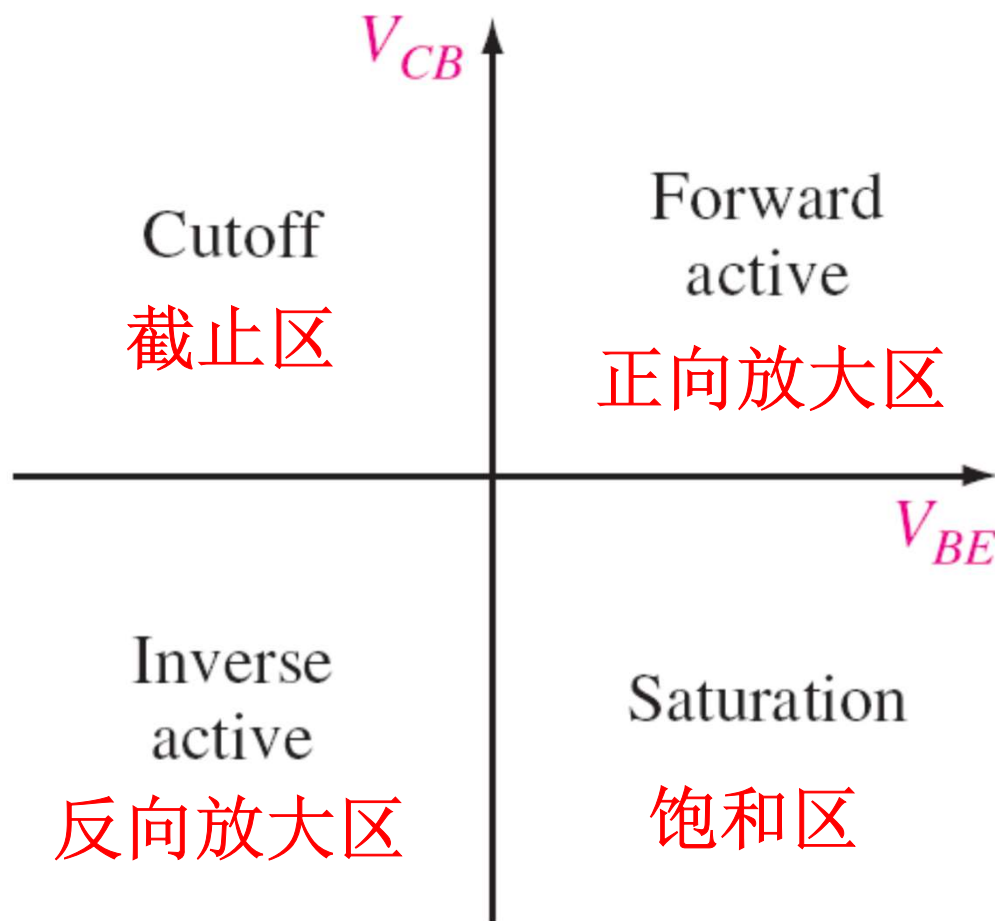
x_B 为电中性基区宽度， L_B 为基区少子扩散长度

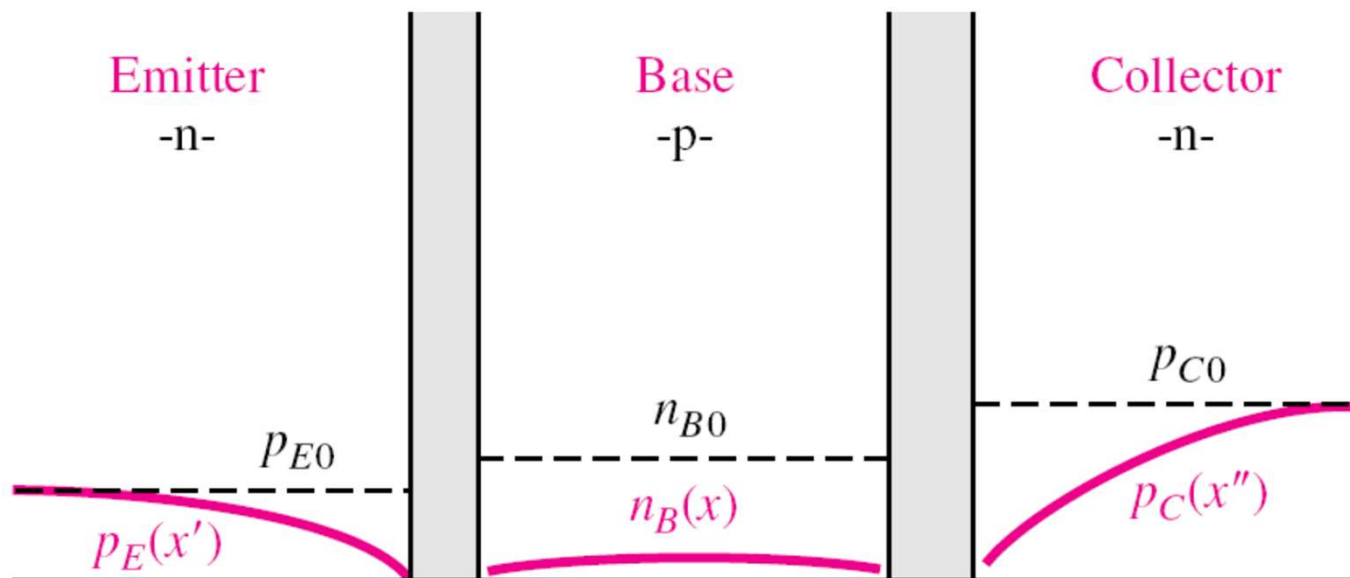


2. 双级晶体管的基本原理

BJT四种工作模式：

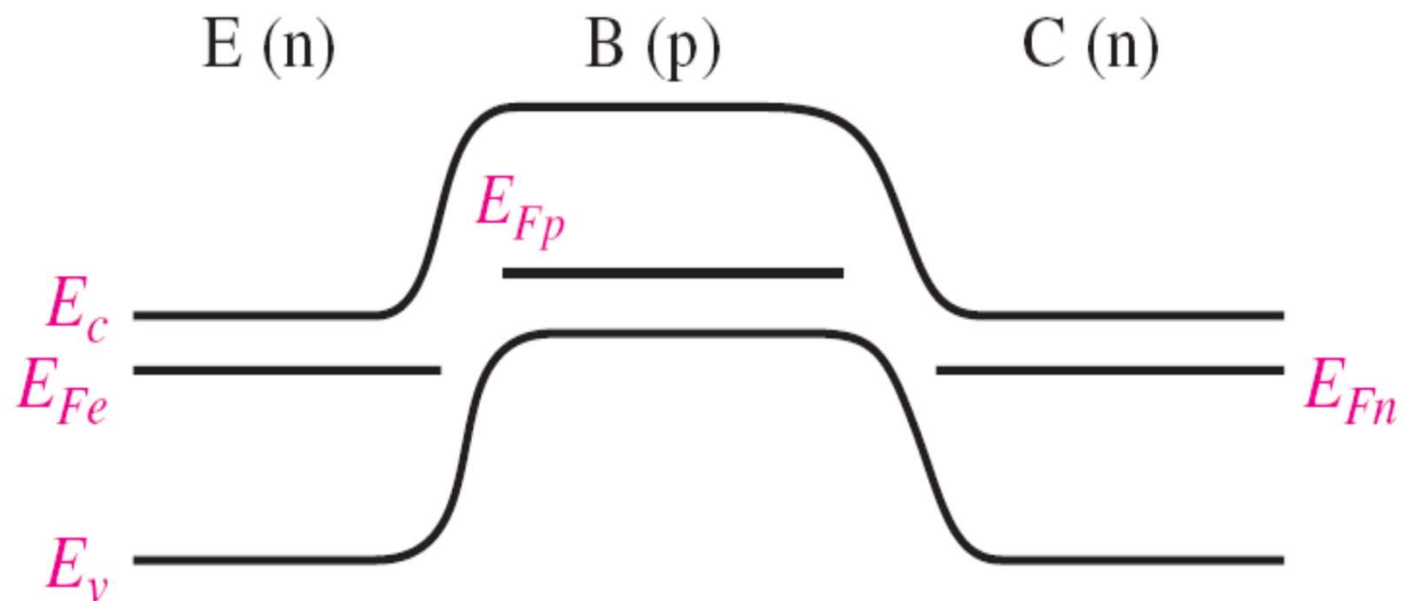
(1) 截止模式； (2) 正向放大模式； (3) 饱和工作模式； (4) 反向放大模式。

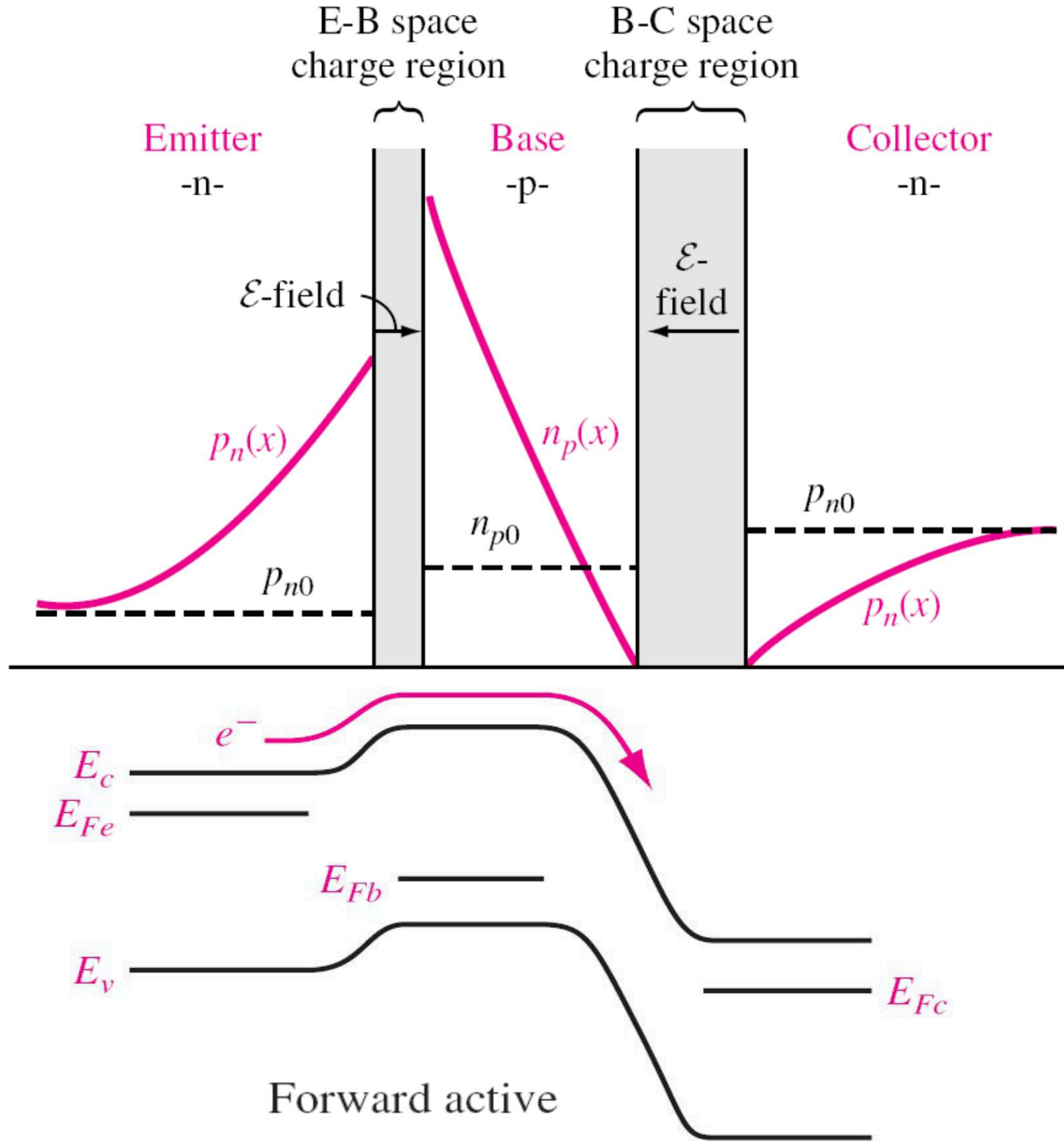




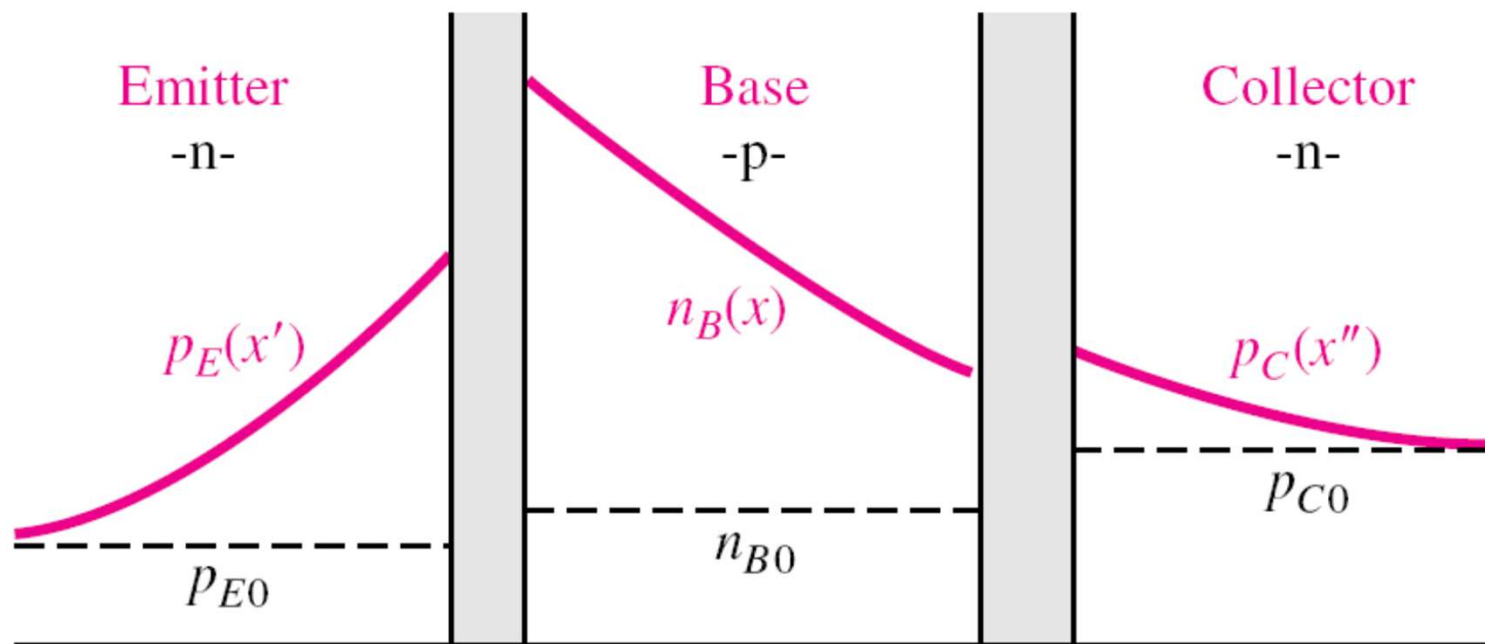
截止区:

发射结反偏，集电结反偏。这两个结的空间电荷区边界处的少数载流子浓度均为零。



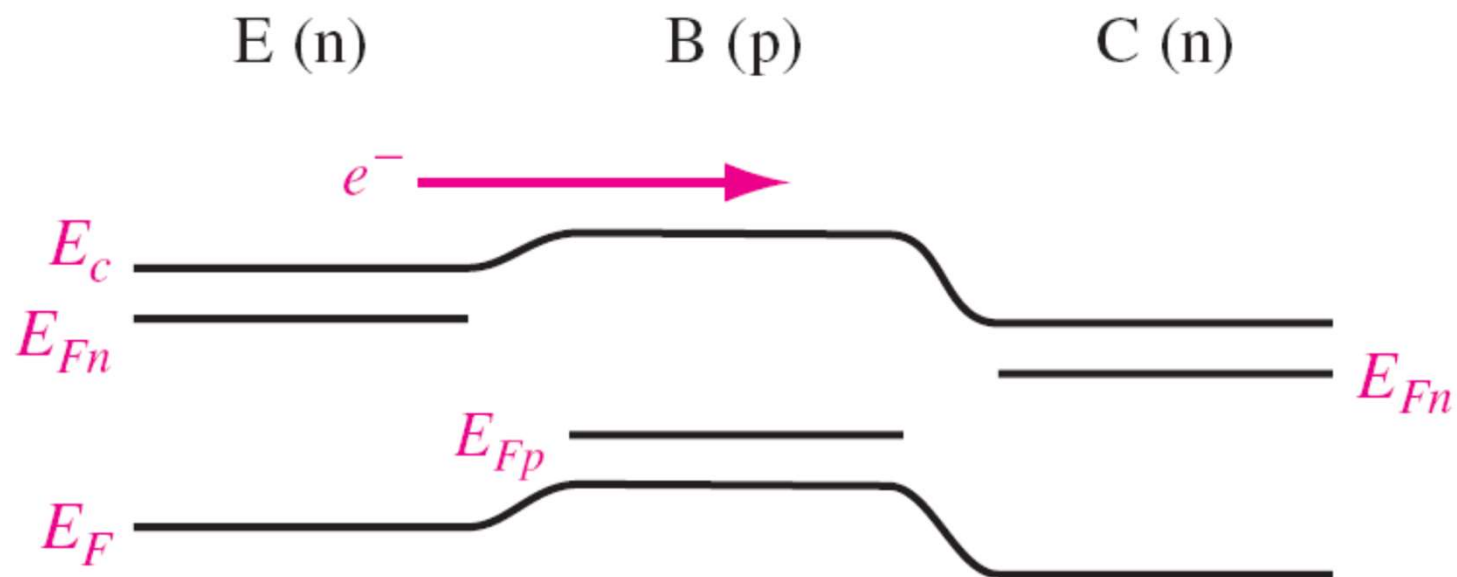


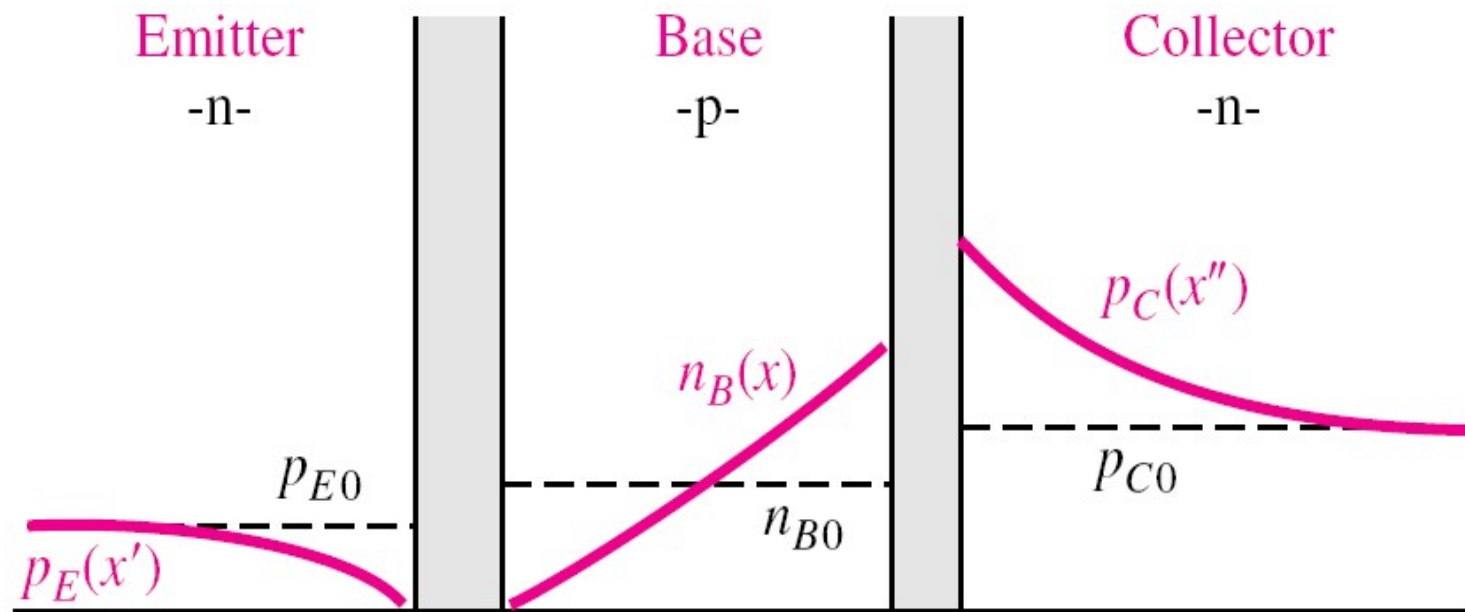
正向放大模式：
发射结正偏，集
电结反偏，内部
各个不同区域少
数载流子的浓度
分布图



饱和区:

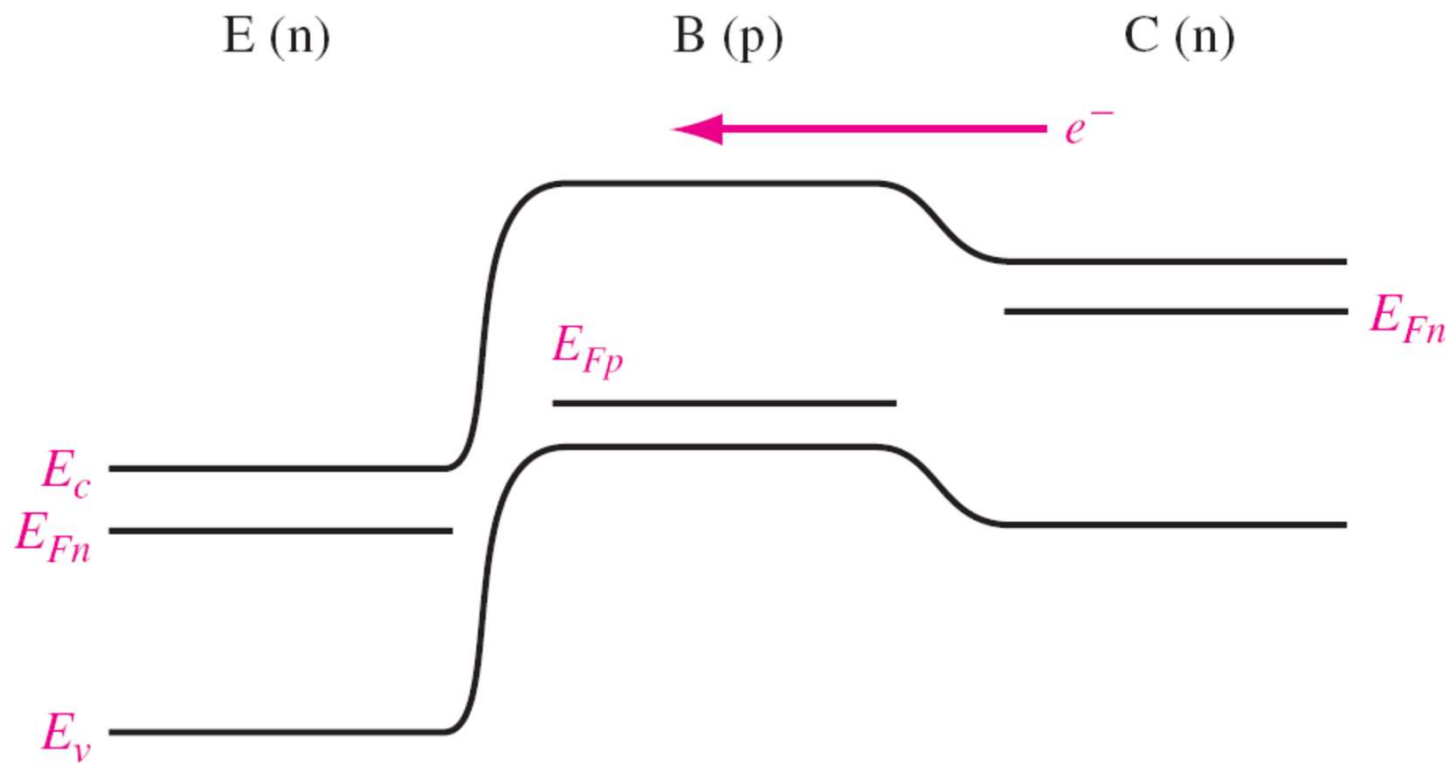
发射结正偏，集电结正偏，内部各个不同区域少数载流子的浓度分布图。





反向放大区:

发射结反偏，集电结正偏，基区中的过剩少子电子的浓度梯度也与正向放大状态正好相反。

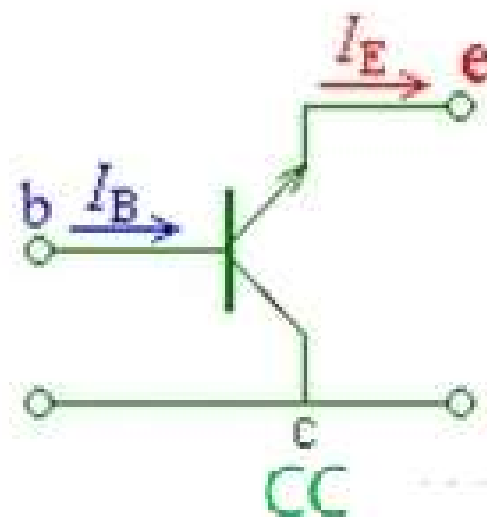
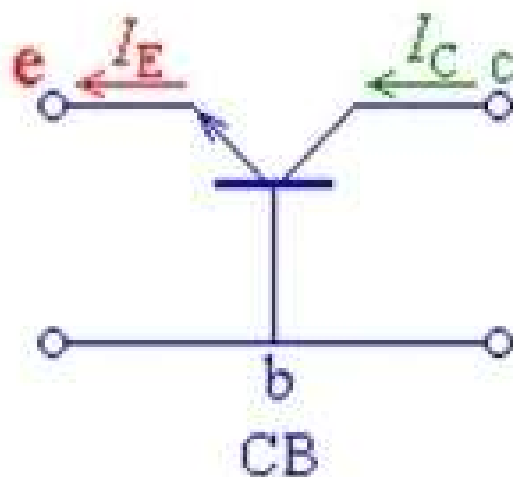
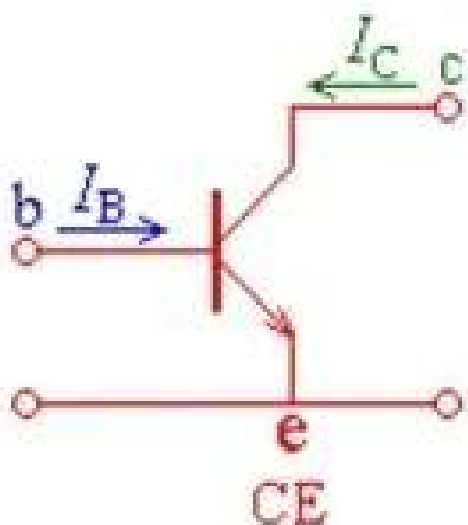


三种连接方式：

共发射极： 电压电路放大，用于一般的信号放大电路，主要利用其电压放大能力来进行信号的电压放大（增益）

共基极： 频率响应范围宽，射频电路应用较多

共集电极： 电流放大,输出阻抗低,带负载能力强，用于功率电路最后一级的功率放大输出。



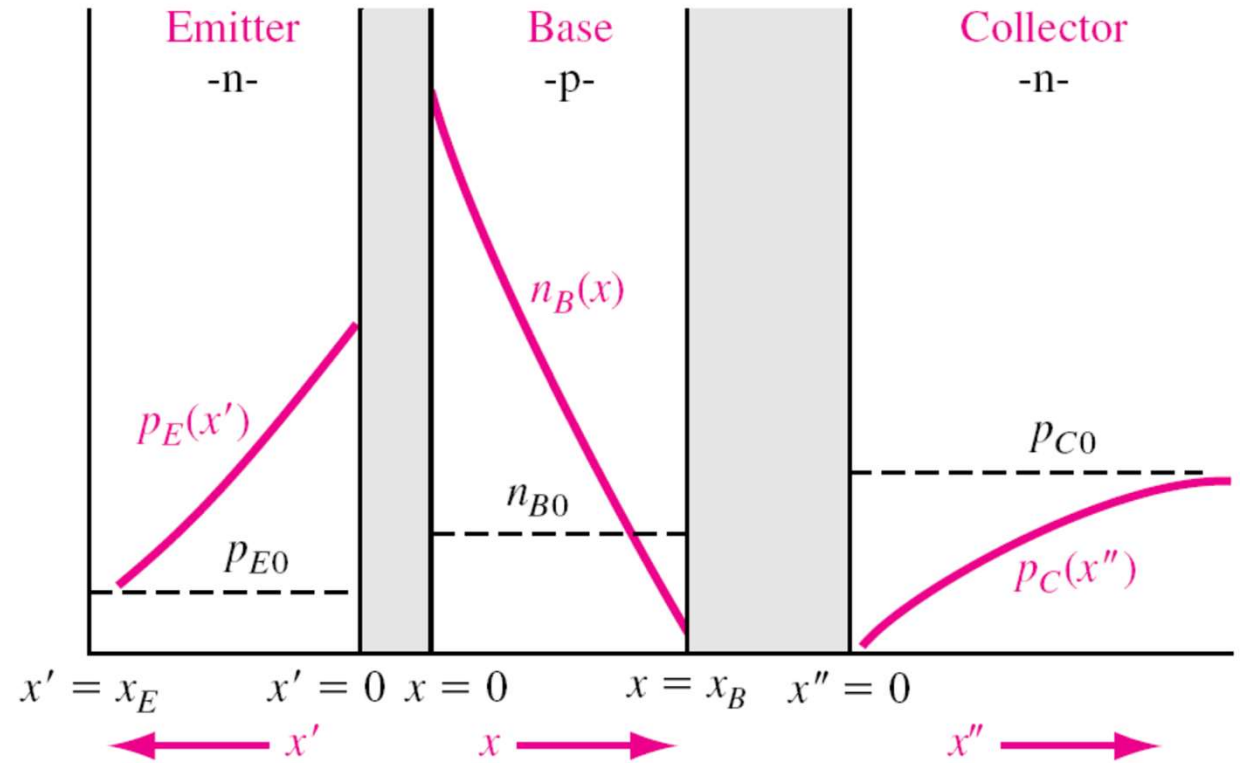
3.过剩少子浓度

$$D \frac{\partial^2 [\delta n(x)]}{\partial x^2} - \frac{\delta n(x)}{\tau_0} = 0$$

$$\delta n(x) = A \exp\left(\frac{x}{L}\right) + B \exp\left(\frac{-x}{L}\right)$$

$$D \frac{\partial^2 [\delta p(x)]}{\partial x^2} - \frac{\delta p(x)}{\tau_0} = 0$$

$$\delta p(x) = C \exp\left(\frac{x}{L}\right) + D \exp\left(\frac{-x}{L}\right)$$



$$D_E \frac{\partial^2 [\delta p_E(x')]}{\partial x'^2} - \frac{\delta p_E(x')}{\tau_{E0}} = 0$$

$$\delta p_E(x_E) = 0$$

$$\delta p_E(0) = p_{E0} \left[\exp\left(\frac{eV_{BE}}{kT}\right) - 1 \right]$$

$$\delta p_E(x') = \frac{p_{E0} \left[\exp\left(\frac{eV_{BE}}{kT}\right) - 1 \right] \sinh\left(\frac{x_E - x'}{L_E}\right)}{\sinh\left(\frac{x_E}{L_E}\right)}$$

$$D_B \frac{\partial^2 [\delta n_B(x)]}{\partial x^2} - \frac{\delta n_B(x)}{\tau_{B0}} = 0$$

$$\delta n_B(x_B) = -n_{B0}$$

$$\delta n_B(0) = n_{B0} \left[\exp\left(\frac{eV_{BE}}{kT}\right) - 1 \right]$$

$$\delta n_B(x) = \frac{n_{B0} \left\{ \left[\exp\left(\frac{eV_{BE}}{kT}\right) - 1 \right] \sinh\left(\frac{x_B - x}{L_B}\right) - \sinh\left(\frac{x}{L_B}\right) \right\}}{\sinh(x_B/L_B)}$$

$$D_C \frac{\partial^2 [\delta p_C(x'')]}{\partial x''^2} - \frac{\delta p_C(x'')}{\tau_{C0}} = 0$$

$$\delta p_C(x'' \rightarrow \infty) = p_{C0}$$

$$\delta p_C(x'' = 0) = -p_{C0}$$

$$\delta p_C(x'') = -p_{C0} \exp\left(\frac{-x''}{L_C}\right)$$

4.双极晶体管的基本参数——电流增益

共基极电流增益 α

- 发射极注入效率因子 γ
- 基区输运因子 α_T
- 复合因子 δ

共射级电流增益 β ,

$$\beta = \frac{\alpha}{1 - \alpha}$$

3.非理想效应

厄立效应——输出电导不为零

大注入效应——电流增益下降

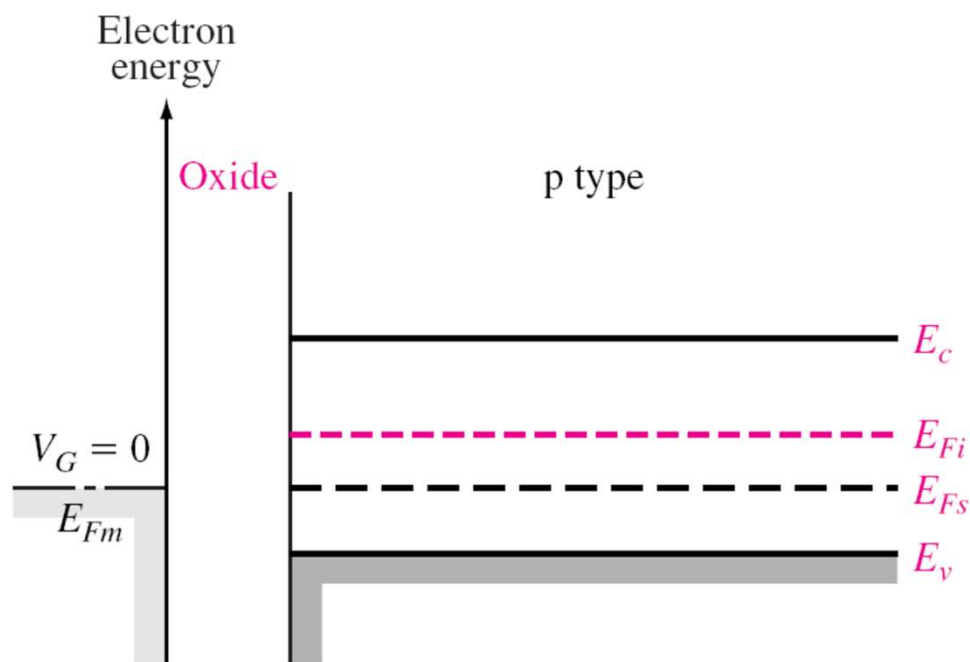
发射区禁带宽度变窄效应——发射结的注入效率降低

发射极电流集边效应——基区电阻的影响

非均匀基区掺杂效应——基区加速场

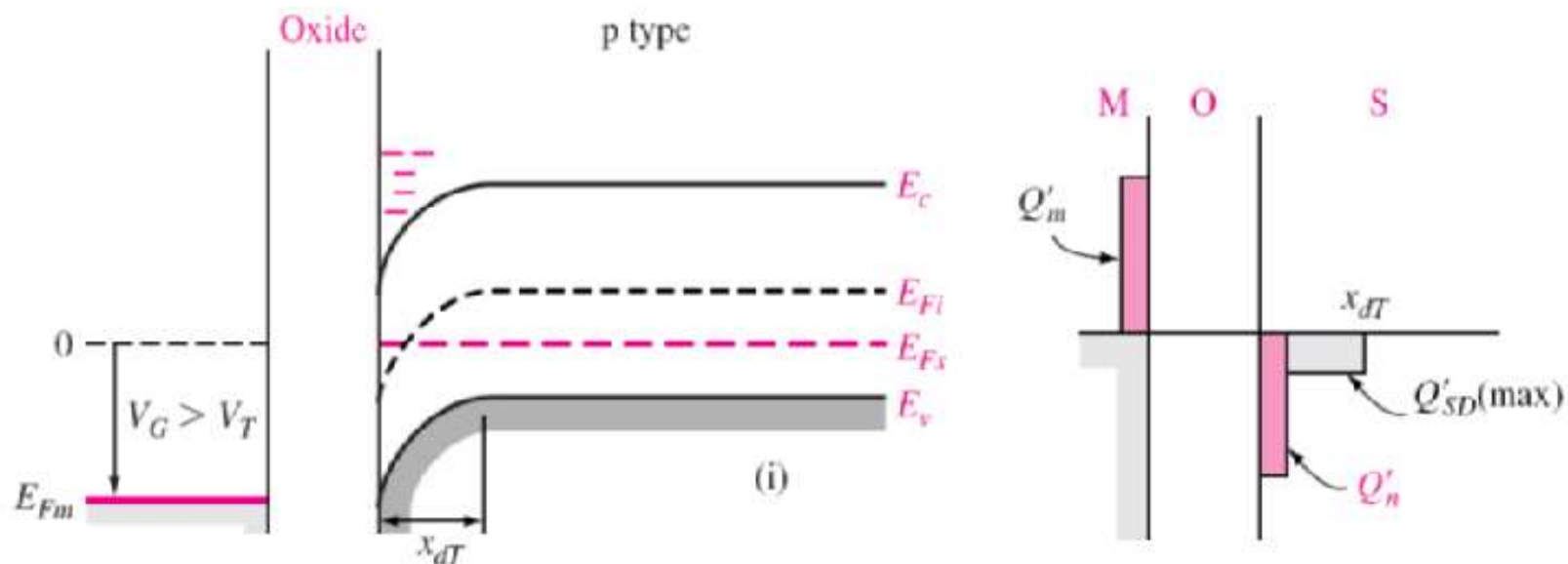
第五章小结

一、理想MOS结构：



当栅极电压为零时，理想情况下的MOS结构能带图为平带，栅电极和半导体材料中的净电荷也都为零。

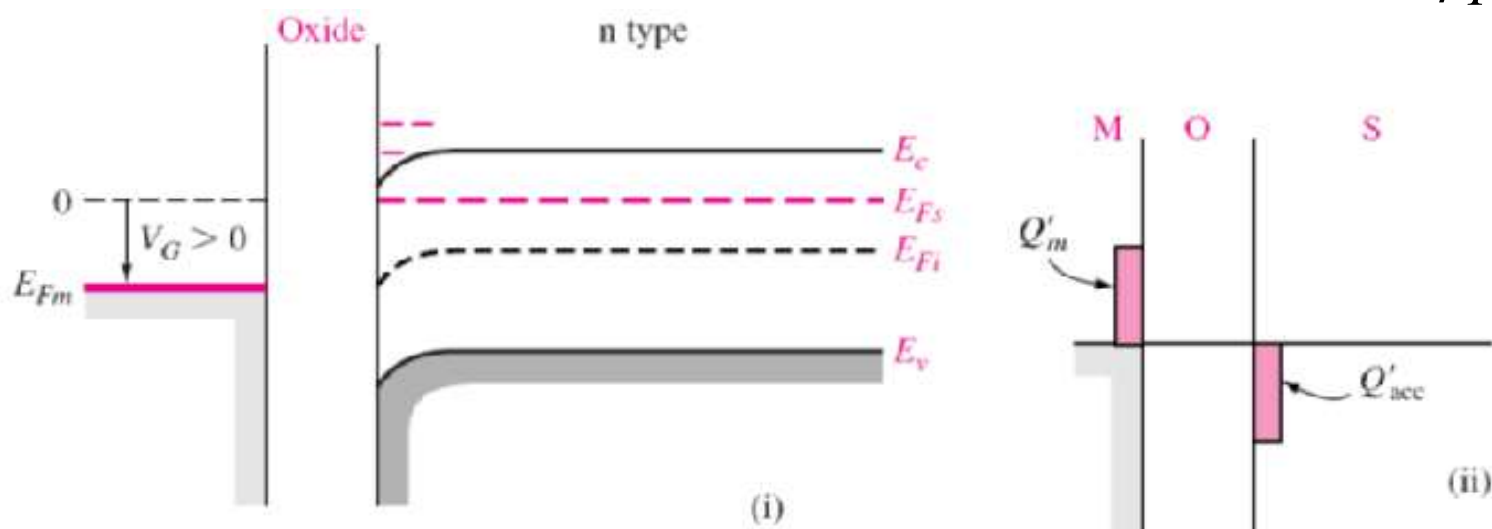


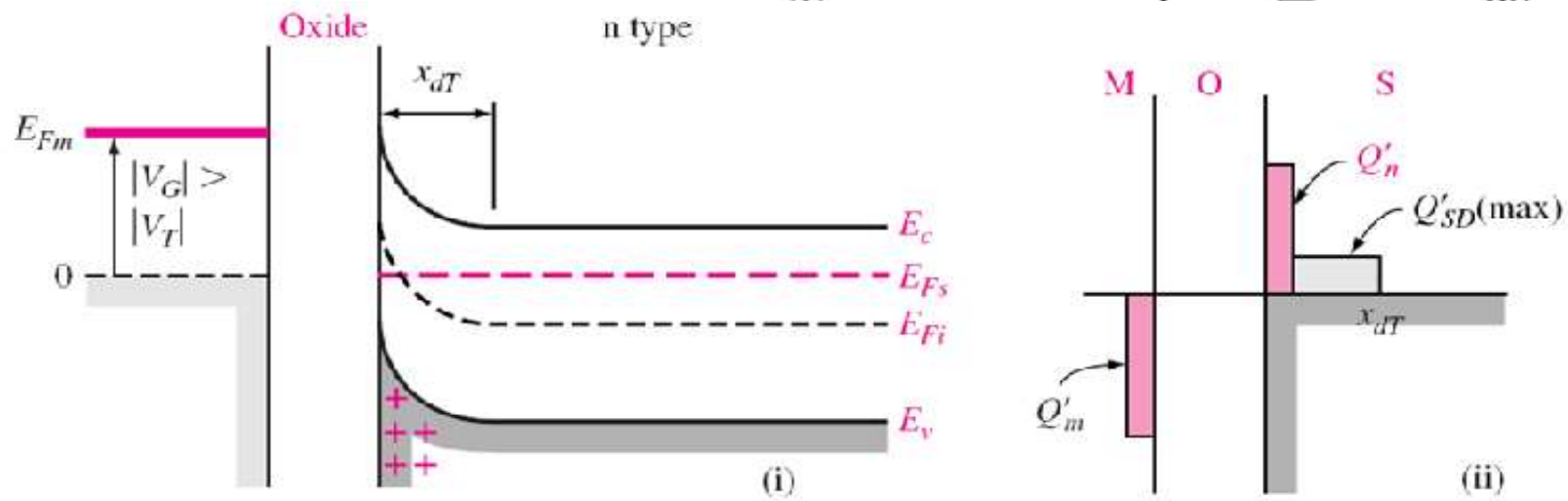
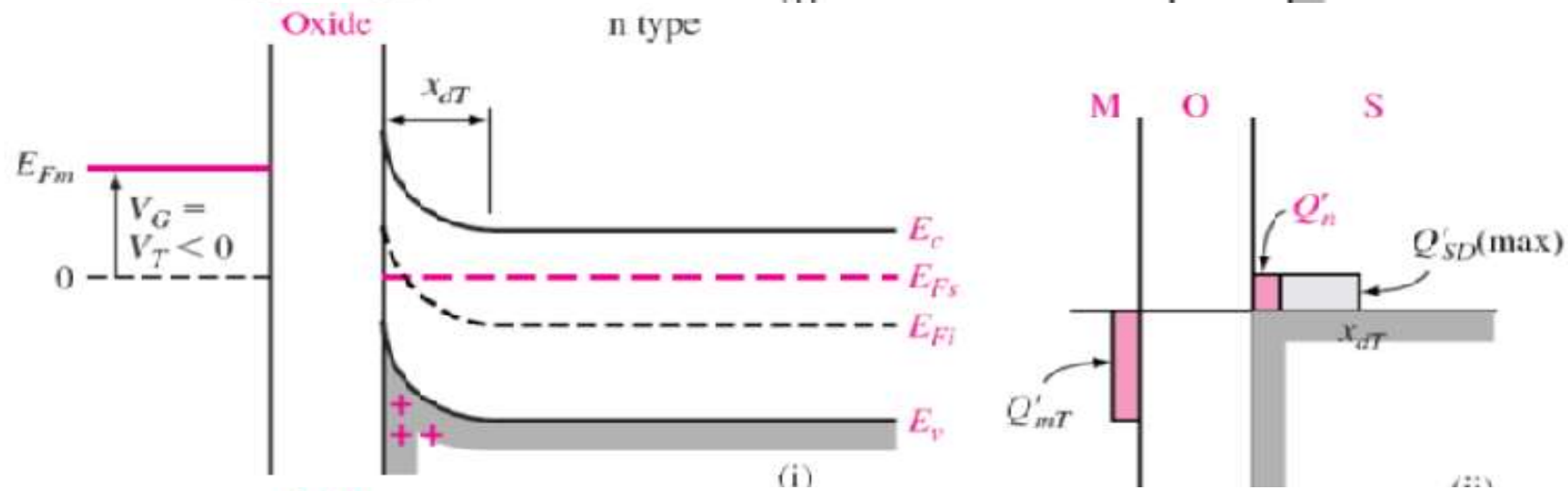
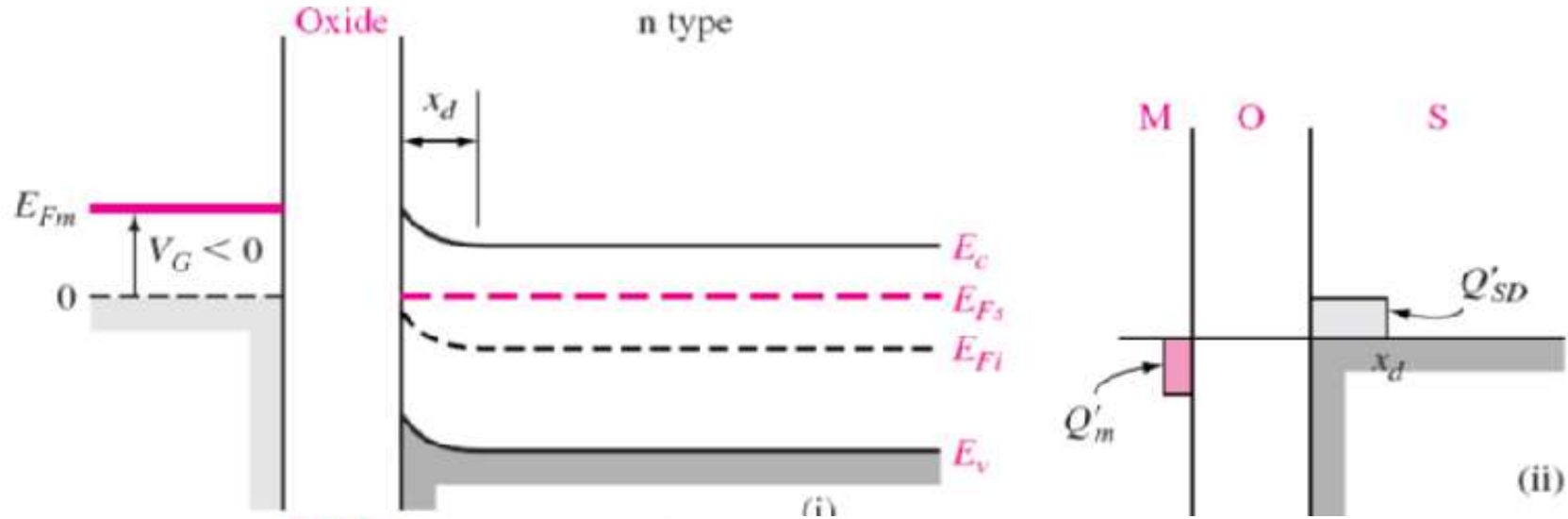


弱反型状态：表面能带往下的弯曲量为 ϕ_{Fp}

强反型状态：表面能带往下的弯曲量为 $2\phi_{Fp}$ ；

当半导体材料表面达到强反型之后，外加的栅电压继续增大，半导体材料的表面势将基本停留在 $2\phi_{Fp}$ ；





$$e\phi_{Fp} = E_F - E_{Fi} = -kT \ln\left(\frac{N_a}{n_i}\right) \quad \phi_s = \phi_{sT} = 2|\phi_{Fp}| \quad x_d = \left(\frac{4\epsilon_s |\phi_{Fp}|}{eN_a}\right)^{1/2}$$

二、实际MOS结构（p型衬底为例）

实际MOS电容中 $\phi_M \neq \phi_S$, 金属-半导体的功函数差

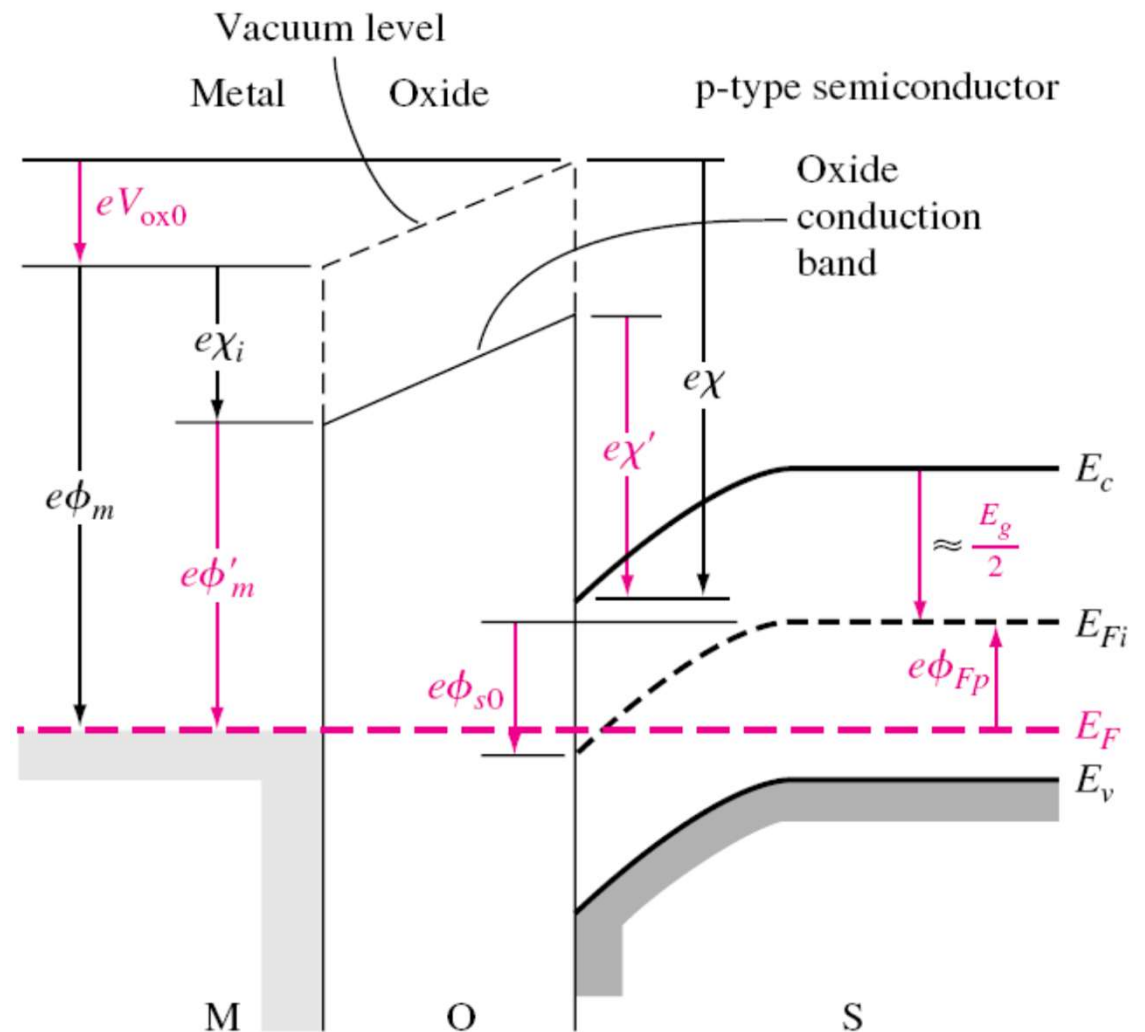
$$\phi_{ms} \equiv [\phi'_m - (\chi' + \frac{E_g}{2e} + |\phi_{Fp}|)]$$

N型多晶硅栅

$$\phi_{ms} = -(\frac{E_g}{2e} + |\phi_{Fp}|)$$

P型多晶硅栅

$$\phi_{ms} = (\frac{E_g}{2e} - |\phi_{Fp}|)$$



平带电压——半导体材料的表面势为零时的栅极电压

$$V_{FB} = \phi_{ms} - \frac{Q'_{ss}}{C_{ox}}$$

第一项抵消功函数差的影响，第二项消除氧化层及界面电荷的影响

阈值电压——MOS电容结构的硅衬底表面发生临界强反型时所需外加的栅极电压。

$$V_{TN} = \frac{Q'_{SD}(\max)}{C_{ox}} - \frac{Q'_{ss}}{C_{ox}} + \phi_{ms} + 2|\phi_{Fp}|$$

$$V_{TN} = \frac{Q'_{SD}(\max)}{C_{ox}} + V_{FB} + 2|\phi_{Fp}|$$

影响阈值电压的三个要素：1) .栅极材料——功函数差；2) .栅氧化层——栅电容和固定正电荷；3) .衬底掺杂浓度

增强型与耗尽型：没有外加栅极电压时，器件的源漏区之间是否已经有导电沟道存在。如果有导电沟道存在，则为耗尽型器件，反之，则为增强型器件。

三、MOS电容的C-V特性

$$C'(acc) = C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

$$\frac{1}{C'(depl)} = \frac{1}{C_{ox}} + \frac{1}{C'_{SD}} \quad C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} \quad C'_{SD} = \frac{\epsilon_s}{x_d} \quad C'_{min} = \frac{\epsilon_{ox}}{t_{ox} + \left(\frac{\epsilon_{ox}}{\epsilon_s}\right)x_{dT}}$$

低频 $C'(inv) = C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$

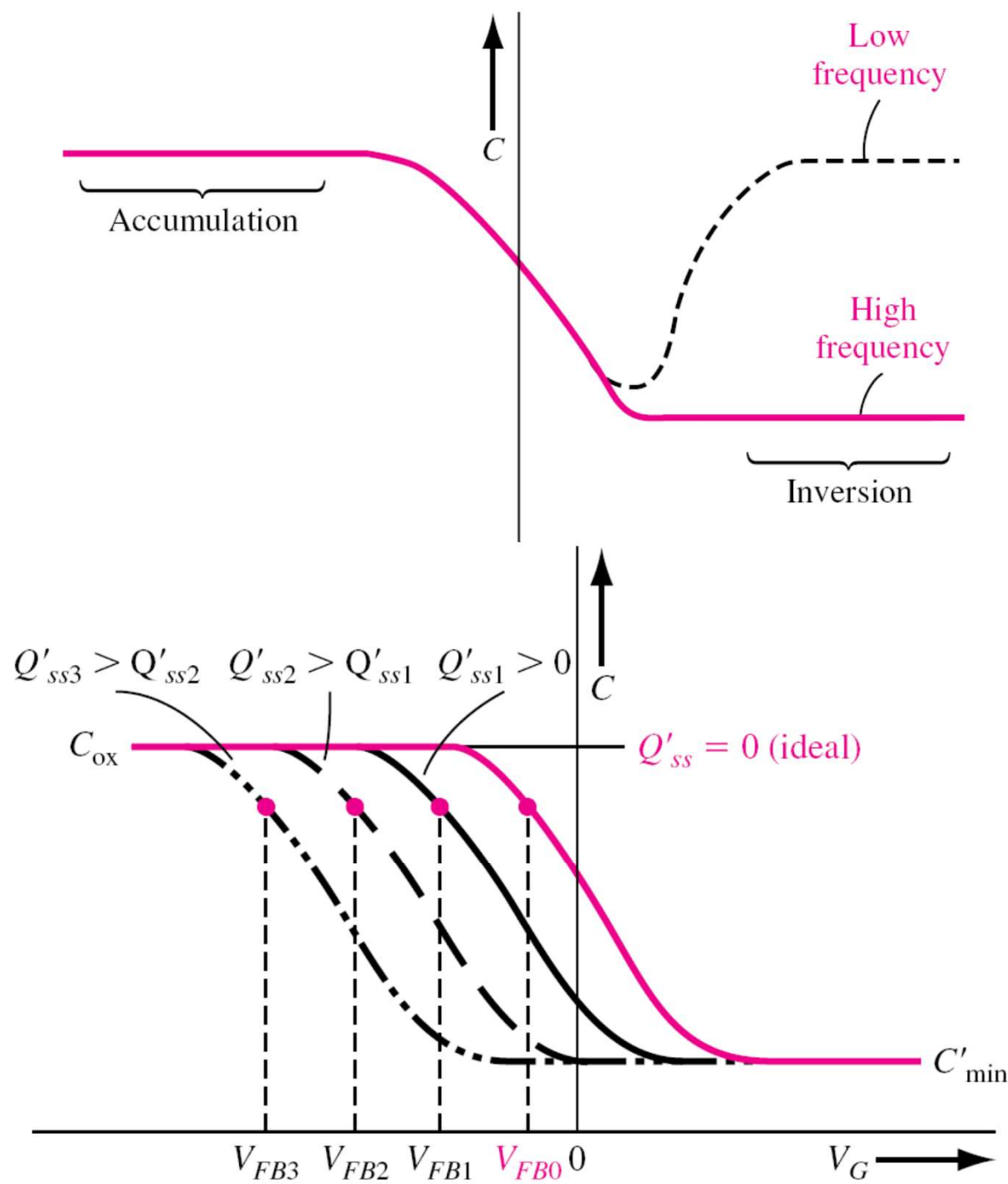
高频 $C'(inv) = C_{min}$

平带电容

$$C'_{FB} = \frac{\epsilon_{ox}}{t_{ox} + \left(\frac{\epsilon_{ox}}{\epsilon_s}\right) \sqrt{\left(\frac{kT}{e}\right) \left(\frac{\epsilon_s}{eN_a}\right)}}$$

$$L_D = \sqrt{\frac{kT\epsilon_s}{e^2 N_a}} \quad \text{德拜长度}$$

高频与低频条件下，p型衬底MOS电容的C-V特性



$$V_{FB} = \phi_{ms} - \frac{Q'_{ss}}{C_{ox}}$$

四、MOSFET的*I-V*特性：基本的物理概念

$$I_D = \begin{cases} 0 & V_{GS} < V_{TN} \\ K_n \left[2(V_{GS} - V_{TN})V_{DS} - V_{DS}^2 \right] & V_{GS} \geq V_{TN} \quad V_{DS} < V_{GS} - V_{TN} \\ K_n (V_{GS} - V_{TN})^2 & V_{GS} \geq V_{TN} \quad V_{DS} \geq V_{GS} - V_{TN} \end{cases}$$

$$K_n = \frac{W}{L} \frac{\mu_n C_{ox}}{2}$$

$$V_{DS(sat)} = V_{GS} - V_{TN}$$

五、MOSFET的非理想效应

1) 亚阈值导电

$$I_D(sub) = I_0 \left[\exp\left(\frac{V_G - V_T}{nV_t}\right) \right] \quad | \phi_{Fp} | < \phi_s < 2 | \phi_{Fp} |$$

$$S \equiv \left. \frac{dV_{GS}}{d(\lg I_{Dsub})} \right|_{0 < V_{GS} < V_T} \approx \frac{kT}{e} \ln(10) = 60mV / dec$$

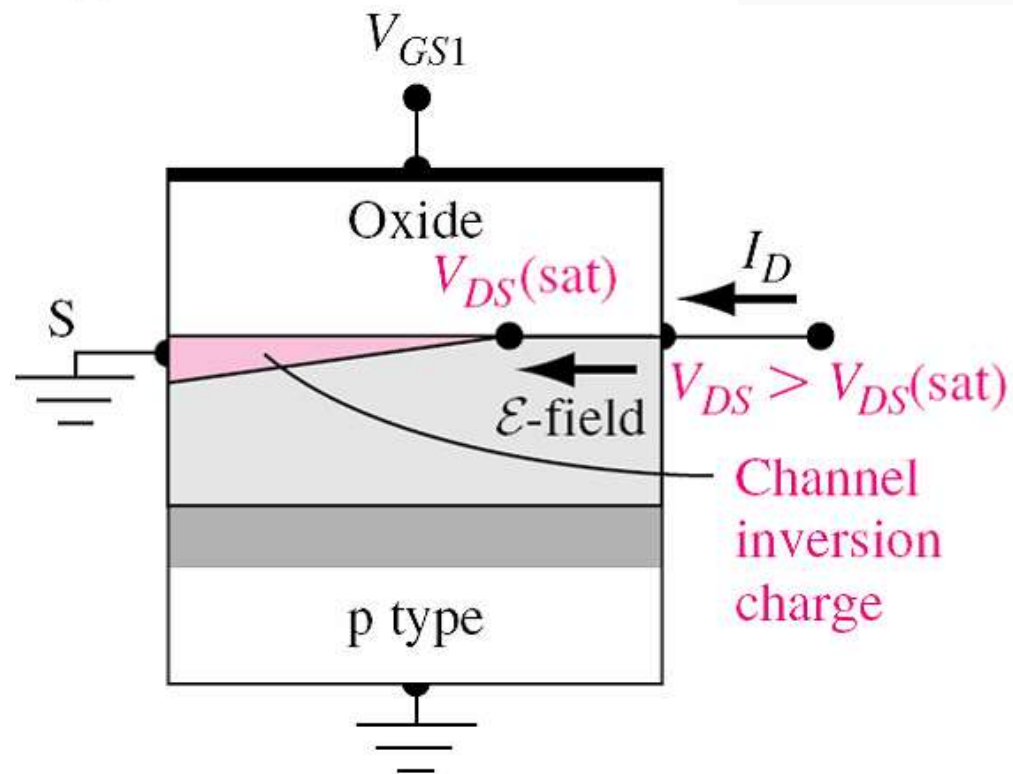
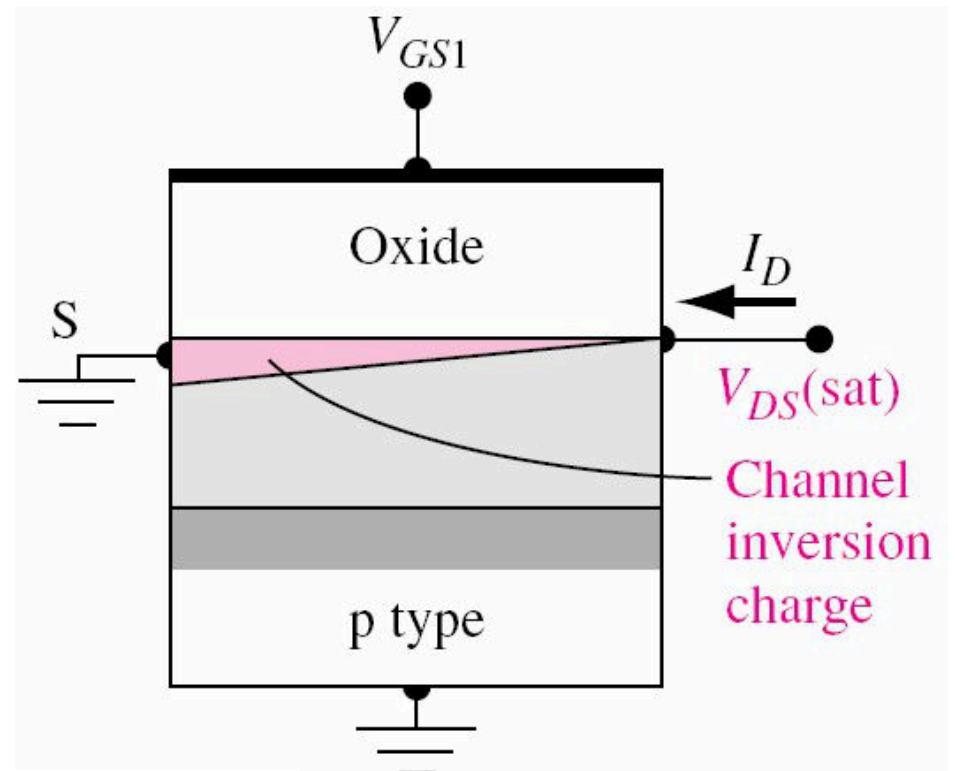
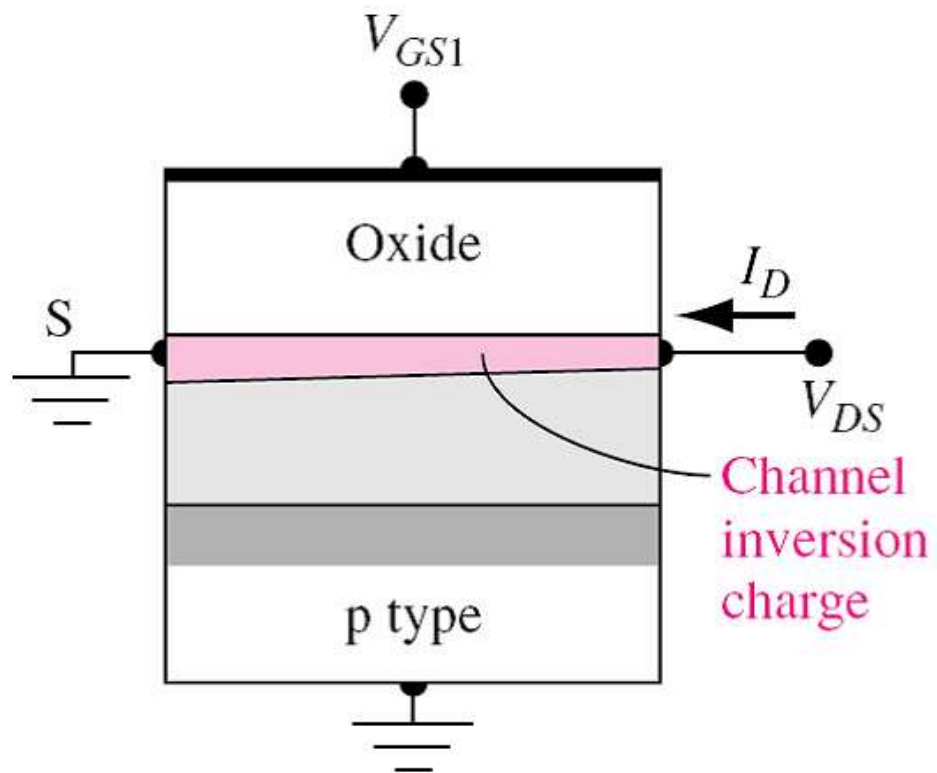
2) 沟道长度调制效应

$$\Delta L = \sqrt{\frac{2\epsilon_s}{eN_a}} \left[\sqrt{V_{bi} + V_{DS}(\text{sat}) + \Delta V_{DS}} - \sqrt{V_{bi} + V_{DS}(\text{sat})} \right]$$

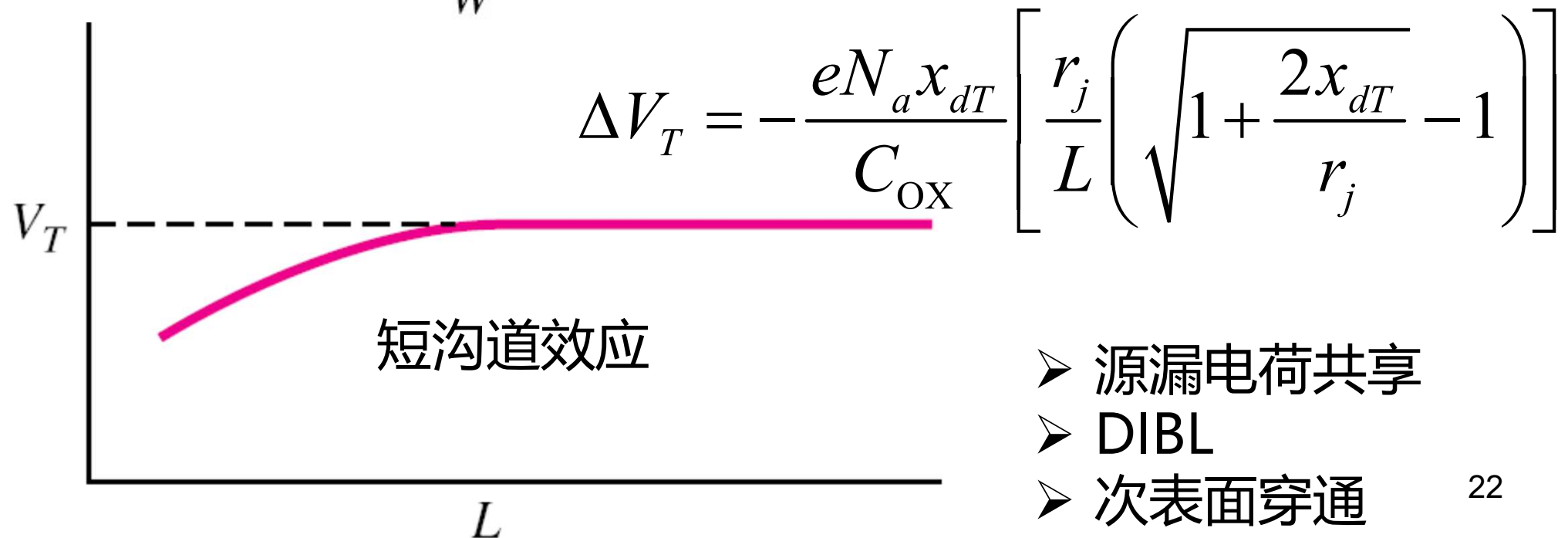
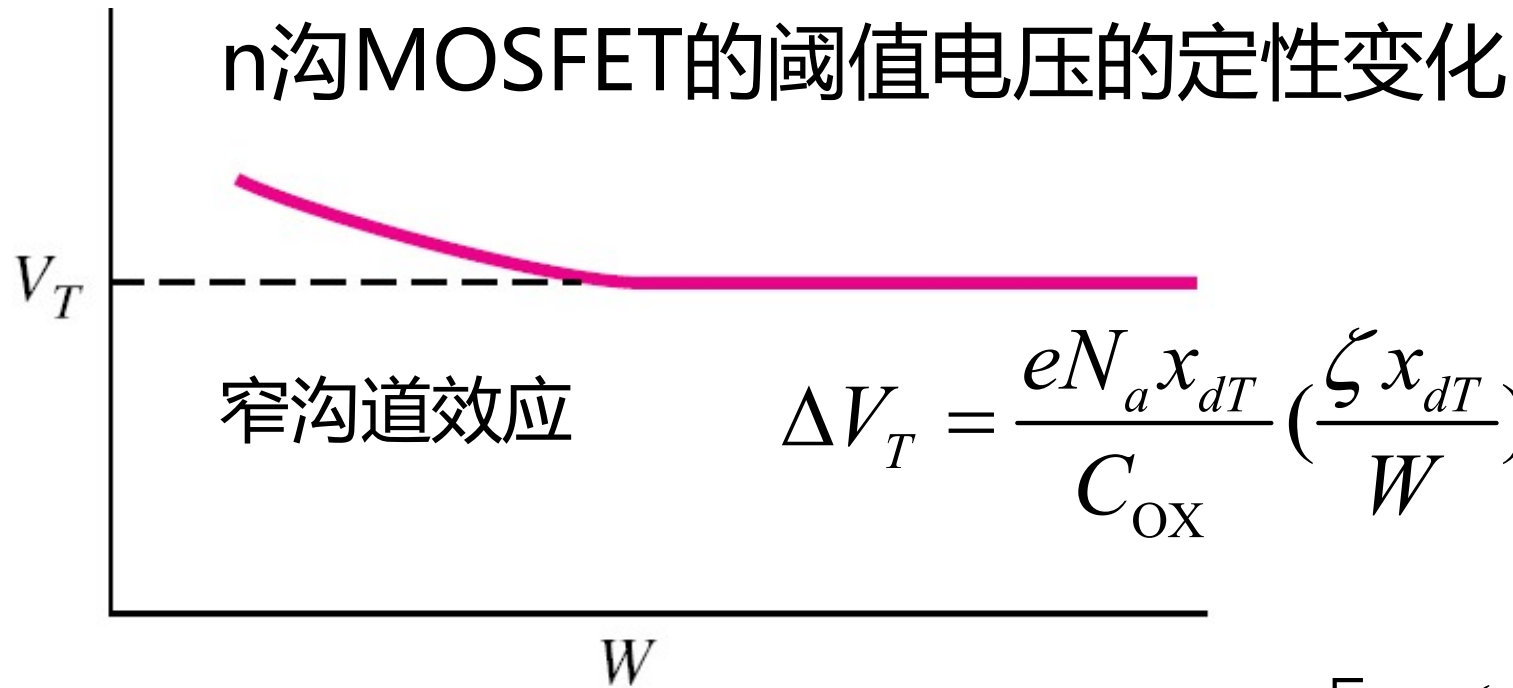
$$\lambda \equiv \frac{\Delta L}{L} \cdot \frac{1}{V_{DS}}$$

$$I'_D = \left[\frac{L}{L - \Delta L} \right] I_D$$

$$I'_D(\text{sat}) = \frac{\mu_n C_{OX} W}{2L} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})^{20}$$



3) 阈值电压的修正——短沟效应和窄沟效应



- 源漏电荷共享
- DIBL
- 次表面穿通

六、MOSFET的等比例缩小

恒定电场的等比例缩小

通常对于两个相邻的工艺技术代来说，比例因子 $k \approx 0.7$

	Device and Circuit Parameters	Scaling Factor ($k < 1$)
Scaled parameters	Device dimensions (L, t_{ox}, W, x_j)	k
	Doping concentration (N_a, N_d)	$1/k$
	Voltages	k
Effect on device parameters	Electric field	1
	Carrier velocity	1
	Depletion widths	k
	Capacitance ($C = \epsilon A/t$)	k
	Drift current	k
Effect on circuit parameters	Device density	$1/k^2$
	Power density	1
	Power dissipation per device ($P = VI$)	k^2
	Circuit delay time ($\approx CV/I$)	k
	Power-delay product ($P\tau$)	k^3